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CENTRAL CAMPUS OF TECHNOLOGY (CCT), HATTISAR, DHARAN

A Project Proposal On
SewaFi: A Web-based Home Services Booking and Dispatch System

Submitted To:
DEPARTMENT OF IT
CENTRAL CAMPUS OF TECHNOLOGY (CCT), HATTISAR, DHARAN, SUNSARI,
KOSHI, NEPAL

*In partial fulfilment of the requirements for the degree of
Bachelor in Information Technology (BIT)*

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February, 2026

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1. Introduction

In many regions of Nepal, particularly outside the Kathmandu Valley, access to reliable home services such as plumbing, electrical repairs, and cleaning remains inefficient and unstructured. Residents often rely on informal networks or personal referrals, resulting in delayed responses and lack of transparency regarding service quality. This lack of a centralized platform for service booking leads to fragmented service delivery, causing significant challenges for both customers and service providers.

To address these challenges, *SewaFi* is proposed as a user-friendly digital platform that connects customers with verified local service providers in real-time. The platform allows users to submit service requests, track progress, and communicate directly with providers, all through a single, transparent system. This reduces uncertainty, eliminates unnecessary follow-ups, and ensures timely service delivery.

From an administrative perspective, *SewaFi* offers a comprehensive dashboard for managing service requests, coordinating workflows, and evaluating provider performance. This centralized system improves efficiency, accountability, and responsiveness by bridging the communication gap between users and service providers.

With the growing adoption of smartphones and the internet in Nepal, *SewaFi* leverages modern web technologies to provide accessible, scalable, and efficient home service solutions. By contributing to the digital transformation of local services, the platform aims to support the delivery of reliable and user-focused services, particularly in underserved districts.

As digital adoption increases in Nepal, with over 39 million mobile connections and 16.5 million internet users, there is a clear opportunity to implement a localized home services booking system that addresses the needs of communities in the region, simplifying access to essential services[1].

2. Problem Statement

Across Nepal, accessing timely and reliable home services such as plumbing, electrical repair, cleaning, and appliance maintenance remains a significant challenge. The existing system is largely informal, relying on personal contacts, word-of-mouth recommendations,

or unorganized local listings. This often results in **delays in service availability, difficulty finding skilled manpower at the required time, lack of transparency in pricing, and uncertainty about service quality and safety** due to the absence of verification mechanisms.

Despite increasing internet and smartphone usage in Nepal, there is still no widely accessible, standardized digital system that efficiently connects households with verified home service providers nationwide. As a result, customers face inconvenience and uncertainty, while many skilled technicians struggle to reach potential clients consistently. This gap highlights the need for a **centralized home service booking and dispatch system** that improves coordination, ensures trust, and enables faster service delivery. **SewaFi** aims to address this problem by providing a structured, digital platform that simplifies service booking, enhances reliability, and bridges the gap between service demand and skilled manpower across the country.

3. Objectives

The platform seeks to achieve the following goals:

1. **To provide secure user management** by allowing customers, service providers, and admins to create accounts, log in, and manage profiles with role-based access controls.
2. **To enable service request creation and tracking** by providing customers the ability to create service requests, track their status, and receive real-time updates.
3. **To automate service provider matching and task assignment** by matching service requests with available providers based on location, service type, and availability, ensuring efficient task allocation.
4. **To enable admin monitoring and system management** by providing administrators with tools to manage users, monitor service requests, and ensure the system operates smoothly and securely.

4. Methodology

4.1. Requirement identification

4.1.1. Study of Existing System / Literature Review

To design **SewaFi**, existing digital home service platforms were reviewed to understand their strengths and limitations. International platforms such as **Urban Company** show how structured online systems can improve service reliability through verified professionals, transparent pricing, and organized booking processes[2]. However, these platforms require significant resources and are difficult to replicate in smaller or developing markets.

In South Asia, **Sheba.xyz** demonstrates the effectiveness of digital marketplaces in connecting households with skilled service providers, highlighting the importance of trust and efficient coordination[3]. In Nepal, **Sajilo Sewa** provides a local example by offering digital home services mainly within the Kathmandu Valley. Its success is driven by provider verification and customer feedback, but its limited geographic coverage leaves many regions underserved[4].

These studies indicate that while existing platforms are effective, they are often not widely accessible or fully localized. **SewaFi** builds on their strengths by offering a **simple, web-based home service booking and dispatch system** designed for broader accessibility and local needs across Nepal.

4.1.2. Requirement Analysis

- **Hardware Specification**

- Dual-core processor (minimum Intel i3 / AMD A4 or similar)
- Minimum 8 GB RAM
- 10–20 GB free (HDD or SSD)
- Router

- **Software Specification**

- Operating system: Windows 10 / Ubuntu 20.04+ / macOS 10.13+
- Front End: React.js, Tailwind CSS

- Back End: Node.js (Express.js)
- Database: PostgreSQL
- IDE: Visual Studio Code

- **Functional Requirements**

- User registration and authentication (Customer, Provider, Admin)
- Service listing and browsing
- Booking creation and tracking
- Provider job acceptance and status updates
- Administrative management of users and services

- **Non-Functional Requirements**

- Usability: Mobile-responsive and easy-to-use interface
- Reliability: Stable performance for prototype usage
- Security: Secure authentication and data protection
- Maintainability: Modular and well-documented codebase

4.2. Feasibility Study

4.2.1. Technical

Developing SewaFi is technically feasible given the availability of required tools and expertise. The system relies on modern, widely supported technologies such as React for front-end development, Node.js with Express.js for back-end logic, and PostgreSQL for data management. Location-based features can be implemented using Leaflet.js with OpenStreetMap data, while secure authentication and validation can be handled using existing libraries. Since all required technologies are mature, open-source, and well-documented, the project can be successfully developed within the defined scope using a capable development team.

4.2.2. Operational

The proposed system is operationally feasible due to the increasing use of internet services and smartphones among local populations. A simple and responsive interface will reduce

barriers to adoption for customers and service providers. The system does not require specialized hardware or complex training, making it suitable for community-level use. Operational processes such as booking, scheduling, and feedback collection are managed through web interfaces accessible from standard devices.

4.2.3. Economic

The development of *SewaFi* is economically feasible as it leverages open-source technologies and cloud infrastructure, minimizing upfront costs. The benefits include improved accessibility to services, increased engagement of local providers and enhanced transparency, which outweigh the development and maintenance expenses. The system's modular design allows for cost-effective updates and future scalability.

Table 4.1: Cost-Benefit Analysis of the Proposed Project

Item	Description	Cost (\$)	Benefit (\$)
Development Costs	Software Development	15,000	-
Hardware Costs	Servers and Equipment	5,000	-
Training Costs	User Training Sessions	2,000	-
Maintenance Costs	Annual Maintenance	1,000	-
Total Costs		23,000	-
Increased Efficiency	Time Savings	-	30,000
Improved User Satisfaction	User Feedback	-	10,000
Revenue Increase	New Customers	-	20,000
Total Benefits		-	60,000
Net Benefit		23,000	37,000

Given the minimal upfront costs, significant expected benefits, and the scalability of the platform, *SewaFi* represents a cost-effective solution for improving access to home services in eastern Nepal. The project has strong economic potential, with a clear path to profitability through increased user engagement and the expansion of services.

4.2.4. Schedule(Gantt chart showing the project timeline)

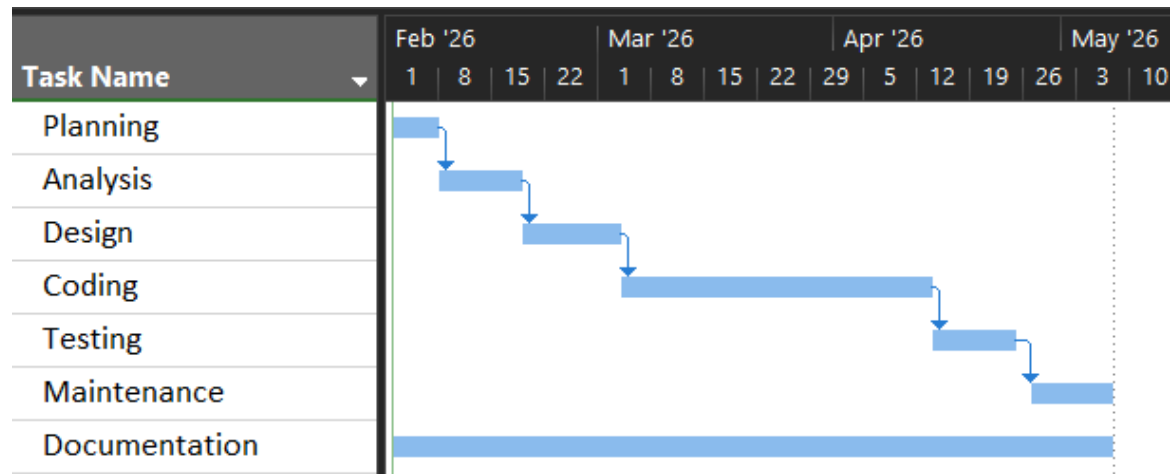


Figure 4.1: Gantt Chart demonstrating schedule feasibility

4.3. High-Level Design of System

4.3.1. Methodology of the proposed system

The SewaFi system will be developed using the Iterative Waterfall Model, allowing for requirements to be refined after the initial prototype phase. This ensures that user feedback from the rural pilot phase can be integrated into the final system.

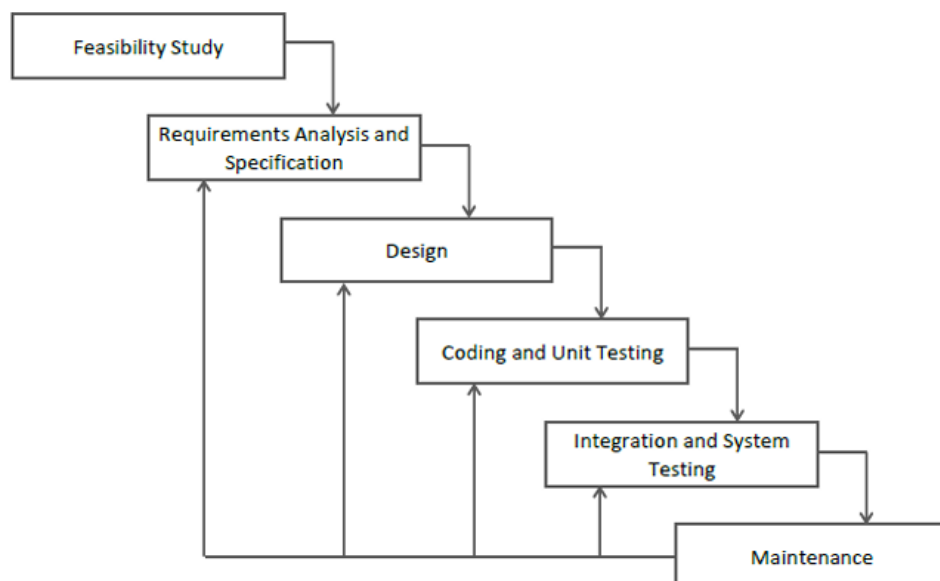


Figure 4.2: Iterative Waterfall Model

4.3.2. Flow Charts/Working Mechanism of Proposed System

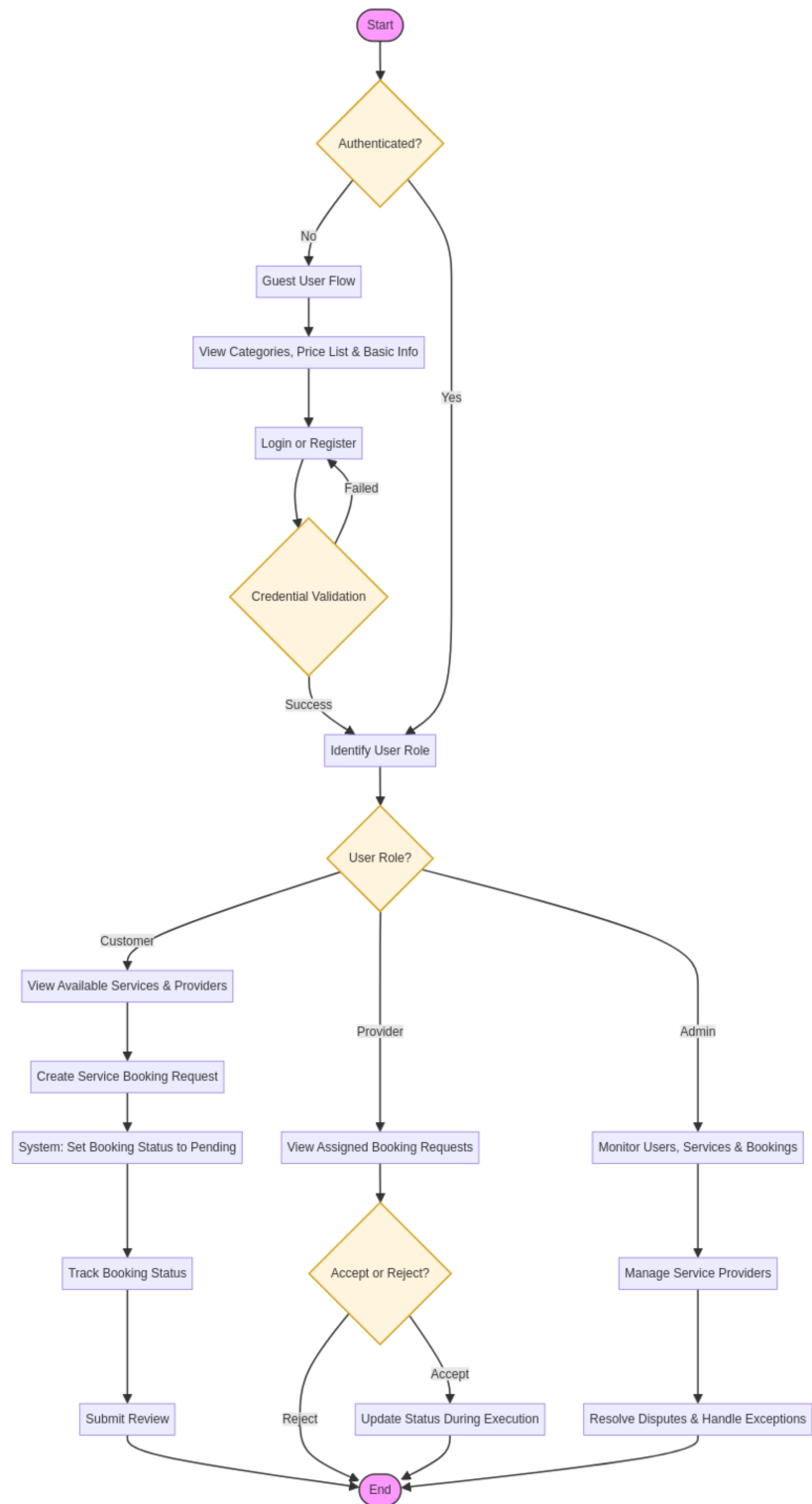


Figure 4.3: Flowchart of the Proposed System

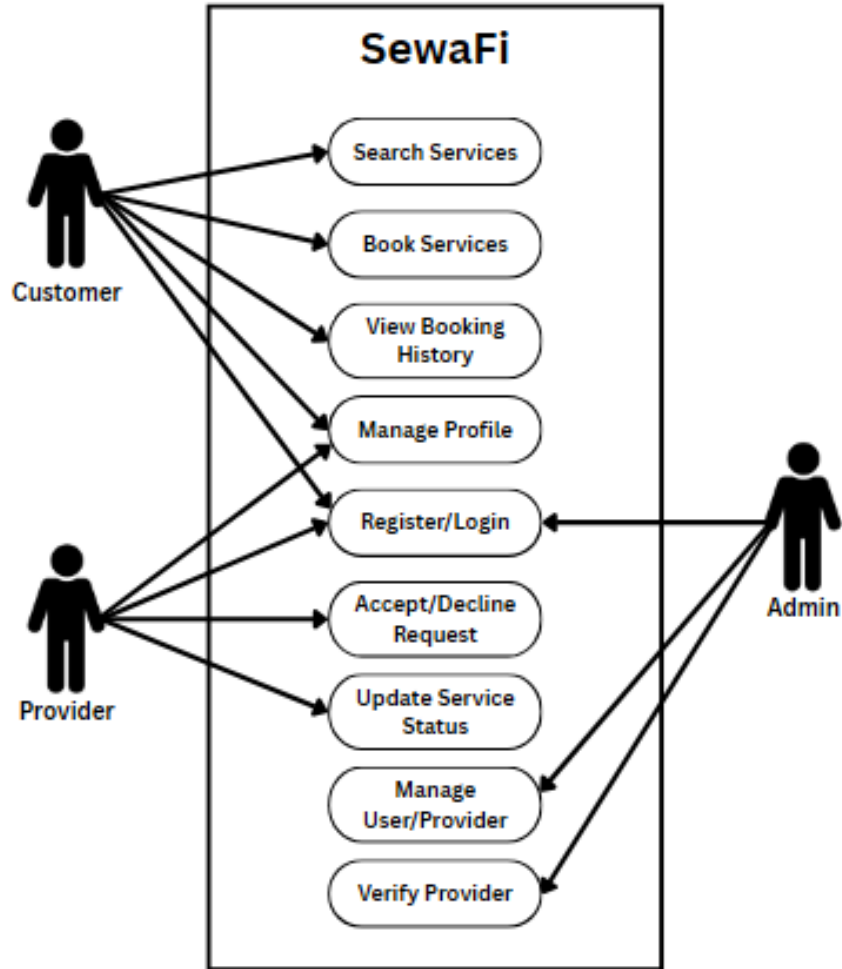


Figure 4.4: Use Case Diagram of the Proposed System

4.3.3. Description of Algorithms

To ensure security, efficiency, and reliability in the *SewaFi* system, several key algorithms and technical mechanisms will be implemented. The major techniques are described below.

1. JSON Web Token (JWT) for Authentication

Purpose: JWT will be used to securely authenticate and authorize users (customers, service providers, and administrators) in a stateless manner, eliminating the need for server-side session storage.

Mechanism: Upon successful login, the system generates a JWT containing the user ID, role, and expiration time. The token is signed using a secure secret key and returned to the client. For subsequent requests, the client includes the token in the request header. The server verifies the token to authenticate the user and enforce access permissions.

Contribution: JWT supports secure, scalable, and role-based access control, ensuring that only authorized users can access specific system functionalities.

2. Bcrypt for Secure Password Storage

Purpose: Bcrypt will be used to protect user passwords stored in the database, preventing disclosure even if the database is compromised.

Mechanism: During registration, passwords are hashed using bcrypt with a randomly generated salt and an appropriate cost factor. Only the hashed password is stored. During login, the entered password is hashed and compared against the stored hash for verification.

Contribution: Bcrypt strengthens system security by preventing plaintext password storage and reducing vulnerability to brute-force attacks.

3. Provider Dispatch and Matching Algorithm

Purpose: This algorithm will efficiently assign customer service requests to suitable providers while minimizing response time.

Mechanism: When a booking is created, the system identifies eligible providers based on service category, location, and availability. Notifications are sent to eligible providers, and the first provider who accepts is assigned using an atomic database update to prevent multiple assignments. If no provider accepts within a predefined time limit, the request is escalated to the administrator for manual handling.

Contribution: This dispatch approach ensures fast and fair service assignment, reduces customer waiting time, and prevents conflicts during provider selection, forming a core part of the booking workflow.

5. Expected Output

The *SewaFi* project will deliver a fully functional web-based prototype capable of:

- Allowing customers to register, browse services, and create bookings.
- Enabling providers to accept and manage service requests.
- Providing an admin dashboard for managing users, services, and bookings.
- Demonstrating the entire workflow from booking creation to service completion.

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